Machine Design: Stress, Materials, Measurement, and Shop Practice

MECHANICAL-ENGINEERING · machine-design-materials-shop-practice · BoardPrep lesson v1

Official examination scope

The PRC August 2026 MELE program identifies Machine Design, Materials and Shop Practice as a distinct subject. This guide teaches selected mechanics and design reasoning; it does not provide current code allowables, certify a component, or cover every manufacturing process.

Learning objectives

- Resolve a force into components and maintain equilibrium.
- Use section modulus for an idealized bending-stress estimate.
- Select a material or manufacturing check using stated requirements and evidence.

1. Forces, moments, and load paths

Begin a machine-element problem with a free-body diagram that shows applied loads, supports, geometry, and axes. Resolve a force into components using the angle’s stated reference direction, then apply force and moment equilibrium. Trace the load through connected parts so the analysis includes the shaft, key, fastener, bearing, and housing that carry the action. A force component has the same units as the original force, and a moment has force times distance. A correct diagram often prevents a sign or geometry error before calculation begins.

2. Stress, fatigue, and stability

For a rectangular section bent about its strong axis, the elastic section modulus is $S = bd^2/6$ and the elementary bending-stress magnitude is = M/S. The result depends on orientation, units, elastic behavior, and a suitable beam model. Repeated or fluctuating loads introduce fatigue questions; slender compression members may need a stability analysis rather than a simple material-strength comparison. A factor of safety or allowable value must come from the applicable design basis, not from an invented universal number. Separate a classroom stress estimate from a complete component design.

3. Materials, tolerances, and manufacturing

Material choice should follow the service environment, loading, temperature, wear, corrosion, manufacturability, and inspection plan. A material label alone does not establish the condition, heat treatment, or measured property of a specific component. Tolerances define acceptable variation; fits relate mating dimensions and assembly function. Select a measurement method with suitable range and resolution, then record the datum, temperature or setup when relevant, and observed value. Manufacturing processes introduce characteristic surface, dimensional, and defect risks that should be inspected against the drawing and acceptance criteria.

4. Quality, maintenance, and shop safety

Reliability and maintenance decisions use failure history, operating conditions, criticality, and inspection evidence. A failure mode helps organize cause and consequence, but a ranking tool is only as good as the team’s evidence and assumptions. Before maintenance or machining, identify stored energy and hazardous motion, isolate equipment under the applicable workplace procedure, and verify the safe state. Shop practice also includes guarding, correct tooling, housekeeping, and material handling. Verify current Philippine occupational-safety rules and machine instructions; this general discussion does not prescribe a site-specific control plan.

Estimate elastic bending stress

Question: A rectangular section is 0.10 m wide and 0.20 m deep under a 2 kN·m bending moment. Using S = bd²/6, estimate the bending stress.
1. Compute S = 0.10(0.20²)/6 = 0.0006667 m³.
2. Convert moment to N·m: M = 2,000 N·m.
3. Compute = M/S = 2,000/0.0006667 3.0 × 10 N/m².
Answer: Approximately 3 MPa under the stated elastic bending model.
Check: The result is a nominal stress estimate; it does not include stress concentrations, fatigue, shear, or design allowables.

Common misconceptions

- A large calculated safety factor makes assumptions irrelevant.
- A nominal material grade proves a component’s installed condition.
- One average stress captures fatigue and local stress concentration.

Practice patterns

- Draw the load path and resolve components before calculating.
- Confirm the section axis and units in a bending calculation.
- Match inspection evidence to the drawing requirement.

Official scope references

Official scope: August 2026 Mechanical Engineers Licensure Examination Program — Professional Regulation Commission; Professional Regulatory Board of Mechanical Engineering — https://www.prc.gov.ph/sites/default/files/exam%20program%20aug%202026%20mele%20signed.pdf

Instructional references

Instructional reference: Introduction to Mechanical Design and Manufacturing — David Jensen; University of Arkansas — https://open.umn.edu/opentextbooks/textbooks/introduction-to-mechanical-design-and-manufacturing
Instructional reference: Manufacturing Processes 4–5 — LamNgeun Virasak; Open Oregon Educational Resources — https://open.umn.edu/opentextbooks/textbooks/645
Instructional reference: University Physics Volume 1 — OpenStax, Rice University — https://openstax.org/details/books/university-physics-volume-1
Instructional reference: NIST/SEMATECH Engineering Statistics Handbook — National Institute of Standards and Technology; SEMATECH — https://www.nist.gov/programs-projects/nistsematech-engineering-statistics-handbook

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